

ServiceNow – SAP S4Hana IDoc

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Scenarios

- In this scenario we will execute several actions from SAP S4Hana IDoc Spoke including Update Material and Create Purchase Order.

Assumptions

- ServiceNow S4Hana IDoc Spoke is installed in the ServiceNow instance.
- Infrastructure: ServiceNow Integration Hub, and SAP S4Hana IDoc are all set-up.

Table of Contents

S. No	Description	Page. No
1.	SAP Configuration S4 Hana IDoc	2
2.	ServiceNow Integration Hub's S4 Hana IDoc Spoke Configuration	3
3.	Test Case 1: Update Material	6
4.	Test Case 2: Create Purchase Order	15

ServiceNow Integration Hub's SAP S4 Hana IDoc Spoke Configuration

Purpose: in the previous section, the SAP S4 Hana IDoc has been configured so that it can be used with S4 Hana IDoc Spoke actions. All actions are performing change in the SAP system through the java connector.

Step 1 – ServiceNow: Create Credential Object

1. Provide Username and password for S4 Hana IDoc endpoint connection. (this will be provided by SAP team)

Basic Auth Credentials
SAP S4 HANA IDOC

Name: SAP S4 HANA IDOC Order: 100

User name: RFC_USER

Password:

Active: ☒

A connection alias resolves your connection and credential at runtime. More than one Credential can be active per Connection Alias at a time. If more than one credential is active, they will be used in order.

Credential alias:

Update Delete

Discovery IP Affinity MID server Search

Credential Affinities					
MID server	Credential ID	IP address	IP address decimal	Domain	Type
No records to display					

Step 2 – ServiceNow: Create Connection in SAP_S4HANA_IDoc connection alias

1. Go to Connection & Credential Aliases
2. Open alias SAP_S4HANA_IDoc
3. Create new connection as following: choose credentials created in previous step, provide host that serves as endpoint for IDoc connection, check use mid server option.

The screenshot shows the 'Connection' configuration page for 'SAP S4 HANA IDOC Connection'. The interface includes a header with navigation icons and 'Update' and 'Delete' buttons. The main form contains several fields: 'Name' (SAP S4 HANA IDOC Connection), 'Active' (checked), 'Domain' (global), 'Credential' (SAP S4 HANA IDOC), 'Connection alias' (sn_sap_s4_idoc_spk.SAP_S4HANA_II), 'Host' (ec2-18-170-34-13.eu-west-2.compute.amazonaws.com), 'Use MID server' (checked), and 'Override default port' (3201). A green box highlights the 'Credential', 'Connection alias', 'Host', and 'Use MID server' fields. A blue box highlights the 'Connection alias' field. Below the main form is the 'Advanced MID Server Configuration' section with tabs for 'Advanced MID Server Configuration' and 'Attributes'. It includes a 'MID Selection' dropdown (Specific MID Server) and a 'MID Server' field (sapmidlnx). At the bottom are 'Update' and 'Delete' buttons.

Connection
SAP S4 HANA IDOC Connection

* Name: SAP S4 HANA IDOC Connection

Credential: SAP S4 HANA IDOC

A connection alias resolves your connection and credential at runtime. Only one Connection is active per Connection Alias at a time. [More Info](#)

* Connection alias: sn_sap_s4_idoc_spk.SAP_S4HANA_II

* Host: ec2-18-170-34-13.eu-west-2.compute.amazonaws.com

Use MID server: ☒

Active: ☒

Domain: global

Override default port: 3201

Advanced MID Server Configuration | Attributes

MID Selection: Specific MID Server

* MID Server: sapmidlnx

Update Delete

Step 3 – ServiceNow: Enable Connection (optional)

If there are multiple connections under the same alias, make sure that newly created connection is active.

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Connection & Credential Aliases

sn_sap_s4_idoc_spk.SAP_S4HANA_IDoc

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This record is in the [SAP S4 HANA IDoc Spoke](#) application, but [Global](#) is the current application. To edit this record click [here](#).

Name

SAP_S4HANA_IDoc

ID

sn_sap_s4_idoc_spk.SAP_S4HANA_IDoc

Application

SAP S4 HANA IDoc Spoke

ⓘ

Type

Connection and Credential

Connection type

Basic

Support Multiple Active Connections

☐

Default Retry Policy

Configuration Template

Select a template for configuring connection and credential records.

Related Links

[Create New Connection & Credential](#)

Connections (1)

Connection Attributes (3)

Child Aliases

≡

🔍

Name

▼

Search

⚙

—

Actions on selected rows...

▼

New

Connection alias = sn_sap_s4_idoc_spk.SAP_S4HANA_IDoc

☐	🔍	Name ▲	Connection type	Active	Credential	Use MID server	Protocol	Host	Override default port	Up
☐		SAP S4 HANA IDOC Connection	Connection	true	SAP S4 HANA IDOC	true		ec2-18-170-34-13.eu-west-2.compute.amazo...	3201	2016

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⏩

1 to 1 of 1

⏪

⏩

Test Case 1: Update Material

ServiceNow will trigger a "update material" request message using Integration Hub's SAP S4Hana IDoc Spoke. The request message will be passed to the S4Hana IDoc interface using the java connector. After the response is retrieved, it will be handled back to ServiceNow.

Step 1 – ServiceNow: Configure Action Update Material

1. Create workflow with name e.g. S4HANA IDOC - PROCUREMENT
2. Add action from S4HANA IDoc Spoke: Update Material
3. Configure action as following and save it

Input Details

- Control Record
 - Sender Port: 8443
 - Partner Number of Sender: S4HCLNT000
- Master Material General Data, Item 1
 - Material: A1234
 - Material Group: 01
 - Base Unit of Measure [MEINS]: EA
 - Lab/Office: 001
 - Weight Unit [GEWEI]: KG
 - Master material short texts (MAKT) > Item 1
 - Material Description: GEARBOX
 - Lang. (ISO 639): ENGLISH
 - Master material C segment (MARC) > Item 1
 - Plant: 9000
 - Purchasing Group: 009
 - MRP Type: PD
 - MRP Controller: 001
 - Planned Deliv. Time: 2
 - GR processing time: 2
 - Period Indicator: M
 - Lot size: EX

- Procurement type: F
- Safety stock: 100
- Availability check: 01
- Master material warehouse/batch segment (MARD) > Item 1
 - Storage Location: ROH1
- Master material material valuation (MBEW) > Item 1
 - Valuation Area: 9000
 - Price control [VPRSV]: V
 - Valuation Class [BKLAS]: 3000
 - Price control [VMVPR]: V
 - Valuation class [VMBKL]: 3000
 - Price control [VJVPR]: V
 - Valuation class [VJBKL]: 3000

1  Update Material 

Action

* Connection Alias    

▼ * Control Record 

* Sender port 

* Partner number of sender 

▼ Master Material General Data  

▼ Item 1  

Function [MSGFN]  

Material  

Material Group	01		
Old material number [BISMT]			
Base Unit of Measure [MEINS]	EA		
Order Unit			
Document			
Document type			
Doc. Version			
Page format [ZEIFO]			
Document change no.			
Page number			
Number of sheets			
Prod./insp. memo			
Page format [FORMT]			
Size/dimensions			
Basic material [WRKST]			
Industry Std Desc.			
Lab/Office	001		
Purchasing value key			
Gross Weight			
Net Weight			
Weight Unit [GEWEI]	KG		

Master material short texts (MAKT)

Drag and drop array.object type pill

Item 1

Drag and drop object type pill

Function

Language Key

Material Description

Lang. (ISO 639)

GEARBOX

ENGLISH

Master material C segment (MARC)

Drag and drop array.object type pill

Item 1

Drag and drop object type pill

Function

Plant

Maintenance status

Deletion Indicator

Valuation Category

ABC Indicator

Critical Part

Purchasing Group

Unit of issue

MRP profile

MRP Type

MRP Controller

Planned Deliv. Time

9000

009

PD

001

2

▼ Master material warehouse/batch segment (MARD)

Drag and drop array.object type pill

+

▼ Item 1

Drag and drop object type pill

⊖

Function

⊖

Storage Location

ROH1

⊖

▼ Master material material valuation (MBEW)

Drag and drop array.object type pill

+

▼ Item 1

Drag and drop object type pill

⊖

Function

⊖

Valuation Area

9000

⊖

Valuation Type

⊖

Deletion Indicator

⊖

Price control [VPRSV]

V

⊖

Moving price [VERPR]

⊖

Standard price [STPRS]

⊖

Price unit [PEINH]

⊖

Valuation Class [BKLAS]

3000

⊖

Price control [VMVPR]

V

⊖

Moving price [VMVER]

⊖

Standard price [VMSTP]

⊖

Price unit [VMPEI]

⊖

Valuation class [VMBKL]

3000

⊖

Price control [VJVPR]

V

⊖

1. Execute workflow

Test Flow

Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.

This trigger type receives no inputs, test flow execution will run immediately on Test.

☐ Run test in background ⓘ

Cancel

Run Test

Step 2 – ServiceNow: Inspect Request Payload

1. After execution is completed, open execution details
2. Open Step 2 execution details and inspect payload



Step 2 - Ping and Post to SAP [Script]

Completed

20

Integration Metadata

Initial MID Server Selection

sapmidlnx

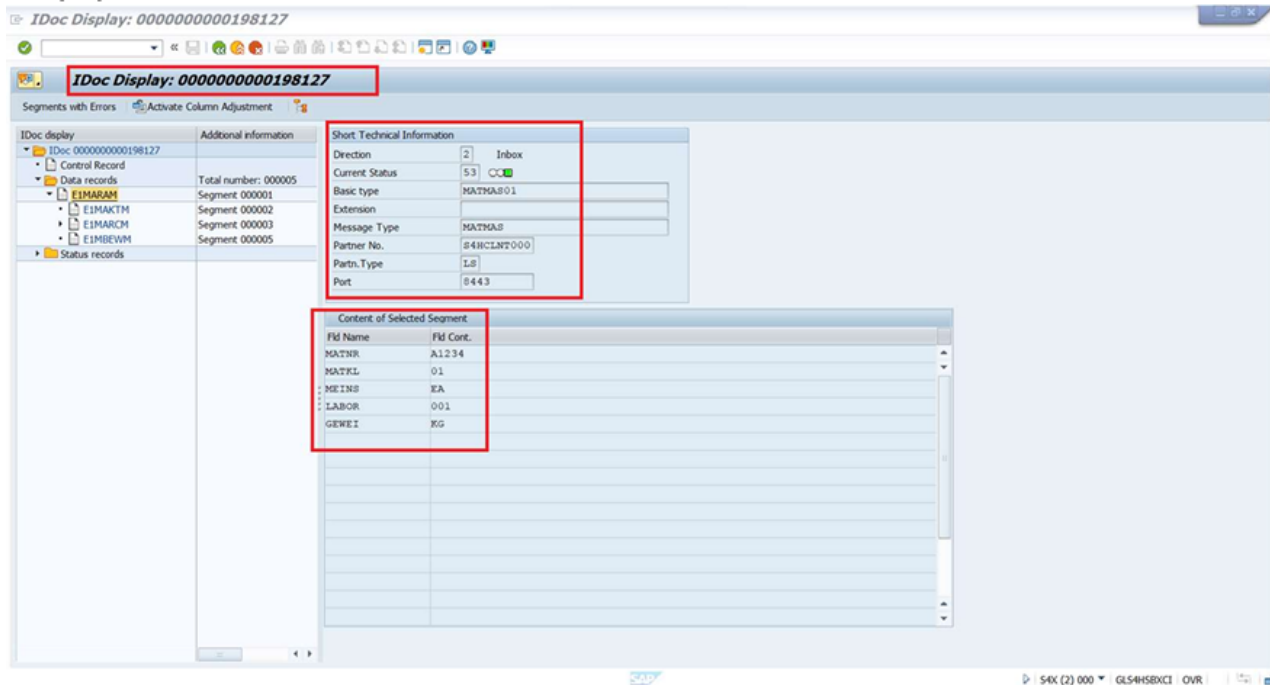
Step Configuration

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_S4HANA_IDoc ⓘ	sys_id ⓘ
Required Runtime	mid	mid
Script	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.commons.cred...	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.commons.cred...
Select MID Server Using	use_connection_alias	use_connection_alias
conn_host	ec2-18-170-34-13.eu-west-2.compute.amazonaws.com	conn_host ⓘ
conn_params	{"u_client":"","u_system_number":"","u_lang":"EN"}	conn_params ⓘ
cred_id	72ddb7ea2f0f311047578d1df699b6d4	cred_id ⓘ
payload	{"IDOC":{"EDI_DC40":{"SNDPOR":"8443","SNDPRN":"S4HCLNT000"},"E1MARAM":{"MFRNR":"","MSTDE":"","_3r_C...	json_in ⓘ

Step Output Data

VARIABLE NAME	RUNTIME VALUE
Step Status	{ "Step Status": { "code": 0, "message": "Success" } }
response	{ "statusCode": "53", "status": "Success", "statusDesc": "Inbound IDoc posted", "tid": "0A001E68852065BFE2EA..." }
status	success

Step 3 – SAP Execution



Step 4 – ServiceNow: Inspect Response Payload

1. Open execution details
2. Inspect output data section – it will contain materials response details

1 **Update Material** Completed

Configuration Details

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_S4HANA_IDoc	SAP_S4HANA_IDoc
Control Record	{ "Control Record": { "SNDPDR": "8443", "SNDPRN": "S4HCLNT000" } }	
Master Material General Data	{ "Master Material General Data": { "AENAM": null, "AENNR": null, "AESZN": null, "ANP": null, "BEGRU": null, "BE...	

Output Data

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Action Status	{ "Action Status": { "code": 0, "message": "Success" } }	
Don't Treat as Error	true	true
Error Message		error_message
Material Response	{ "Material Response": { "docnum": "0000000000199018", "status": "Success", "statusCode": "53", "statusDesc": "...	
Status	success	status

No Logs

Object can be further inspected by clicking on value.

Viewing response [object]



Raw Text

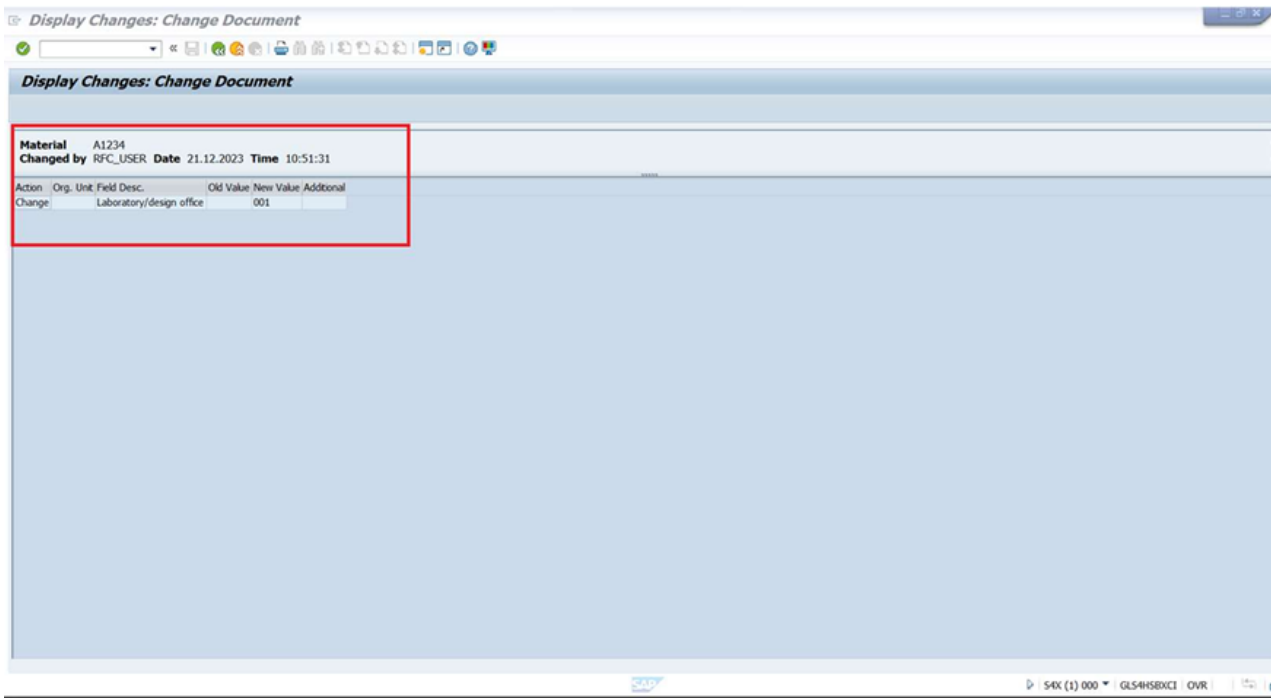
Formatted

```
{
  "Material Response": {
    "docnum": "0000000000199018",
    "status": "Success",
    "statusCode": "53",
    "statusDesc": "Inbound IDoc posted",
    "tid": "0A001E68852065BFE2EA0002"
  }
}
```

Close

Step 5 – Validation on SAP Side

Display Material changes using TCODE MM04



Test Case 2: Create Purchase Order

ServiceNow will trigger an “create purchase order” request message using Integration Hub's SAP S4 Hana IDoc Spoke. The request message will be passed to the S4 Hana IDoc interface using the java connector. After the response is retrieved, it will be handled back to ServiceNow.

Step 1 – ServiceNow: Configure Action Create Purchase Order

1. Create workflow with name e.g. S4HANA IDOC - PROCUREMENT
2. Add action from S4 Hana IDoc Spoke: Create Purchase Order
3. Configure action as following and save it

Input Details

- Control Record
 - Sender Port: 8443
 - Partner Number of Sender: S4HCLNT000
- PO Header
 - Document Date: 20231221
 - Purchasing Doc. Type: NB
 - Purch. Doc. Category: F
 - Company Code: 9000
 - Purch. Organization: 9000
 - Purchasing Group: 009
 - Vendor: 0000100003
- PO Lines > Line 1
 - Item [PO_ITEM]: 00010
 - Material [PUR_MAT]: AA123
 - Storage Location: ROH1
 - Material Group: 01
 - Short Text: CLUTCH PLATE
 - Plant: 9000
 - Order Unit: EA
 - Net order price: 100
 - Price unit: 1
 - Order Price Unit: EA
 - Tax Code: V0
 - Line Items Delivery Schedule > Delivery Schedule 1
 - Item: 00010
 - Delivery date: 20231221
 - Scheduled Quantity: 100
- Addresses for Inbound Delivery > Address 1
 - Item: 00010

- Name: HITACHI PLANT
- City: BANGALORE
- Postal Code: 600009

2



Create Purchase Order ⓘ

Action

Create Purchase Order ▼

* Connection Alias

sn_sap_s4_idoc_spk.SAP_S4HANA_IDoc X▼ ⓘ ↗ ⌵



* Control Record

Drag and drop object type pill ⓘ

* Sender port

8443 ⓘ

* Partner number of sender

S4HCLNT000 ⓘ

▼ PO Header

Drag and drop object type pill ⓘ

Document Date

20231221 ⓘ

* Purchasing Doc. Type

NB ⓘ

Purch. Doc. Category

F ⓘ

* Company Code

9000 ⓘ

* Purch. Organization

9000 ⓘ

* Purchasing Group

009 ⓘ

Outline agreement

* Vendor

0000100003 ⓘ

▼ * PO Lines

Item [AT_ITEM]

Order Unit

EA

Net order price

100

Price unit

1

Quantity Conversion [CONV_NUM1]

Quantity Conversion [CONV_DEN1]

Order Price Unit

EA

Line Items

Delivery Schedules

Drag and drop array.object type pill

Delivery Schedule 1

Drag and drop object type pill

Item

00010

Schedule Line

Deliv. date category [DEL_DATCAT]

Delivery date

20231221

Time

Scheduled Quantity

100

▼ Addresses for Inbound Delivery

Drag and drop array.object type pill

Item 1 Drag and drop object type pill

Item 00010

Address number

Form of address

Name HITACHI PALNT

Name 2

Name 3

Name 4

c/o

City Bangalore

District

City Code [CITY_NO]

Postal Code 600009

1. Execute workflow

Test Flow ×

Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.

This trigger type receives no inputs, test flow execution will run immediately on Test.

☐ Run test in background ⓘ

Cancel Run Test

Step 2 – ServiceNow: Inspect Request Payload

1. After execution is completed, open execution details
2. Open Step 2 execution details and inspect payload

 Step 2 - Ping and Post to SAP [Script] Completed 202

Integration Metadata

Initial MID Server Selection	sapmidlnx	
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Step Configuration

VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_S4HANA_IDoc @	sys_id
Required Runtime	mid	mid
Script	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.commons.cred...	(function execute(inputs, outputs) { try { var credProvFactory = Packages.com.snc.commons.cred...
Select MID Server Using	use_connection_alias	use_connection_alias
conn_host	ec2-18-170-34-13.eu-west-2.compute.amazonaws.com	conn_host
conn_params	{"u_client":"000","u_system_number":"00","u_lang":"EN"}	conn_params
cred_id	72ddb7ea2f0f311047578d1df699b6d4	cred_id
payload	{ "IDOC": { "EDI_DC40": { "SNDPDR": "8443", "SNDPRN": "S4HCLNT000", "E1PORDCR": { "E1BPEKKOC": { "CREATED_BY": "" } } } } }	json_in

Step 3 – SAP Execution

IDoc Display: 000000000198127

Segments with Errors Activate Column Adjustment

IDoc display	Additional information	Short Technical Information
- IDoc 000000000198127 - Control Record - Data records - E1MARM Segment 000001 - E1MAKTM Segment 000002 - E1MARCM Segment 000003 - E1MBEWM Segment 000005 - Status records	Total number: 000005 Segment 000001 Segment 000002 Segment 000003 Segment 000005	Direction: 2 Inbox Current Status: 53 CC Basic type: MATMAS01 Extension: Message Type: MATMAS Partner No.: S4HCLNT000 Partn.Type: LB Port: 8443

Content of Selected Segment

Fld Name	Fld Cont.
MATNR	A1234
MATKL	01
MEINS	EA
LABOR	001
GEWEI	KG

SAP

S4X (2) 000 GLS4HSBICI OVR

Step 4 – ServiceNow: Inspect Response Payload

1. Open execution details
2. Inspect output data section – it will contain materials response details

2  **Create Purchase Order**

Configuration Details		
VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Connection Alias	SAP_S4HANA_IDoc ⓘ	SAP_S4HANA_IDoc ⓘ
Control Record	{ "Control Record": { "SNDPOR": "8443", "SNDPRN": "S4HCLNT000" } }	
PO Header	{ "PO Header": { "AGREEMENT": null, "CO_CODE": "9000", "CREATED_BY": null, "DOC_CAT": "F", "DOC_DATE": "20231221..." } }	
PO Lines	{ "PO Lines": { { "ACCTASSCAT": null, "ACKNOWL_NO": null, "ACKN_REQD": null, "ADDRESS": null, "AGMT_ITEM": null, "..." } } }	
PO Account Assignments	{ "PO Account Assignments": { { "ASSET_NO": null, "BUS_AREA": null, "COST_CTR": null, "COST_OBJ": null, "CO_AREA": null, "..." } } }	
Addresses for Inbound Delivery	{ "Addresses for Inbound Delivery": { { "ADDR_NO": null, "ADR_NOTES": null, "BUILDING": null, "CHKSTATUS": null, "..." } } }	

Output Data		
VARIABLE NAME	RUNTIME VALUE	CONFIGURATION
Action Status	{ "Action Status": { "code": 0, "message": "Success" } }	
Don't Treat as Error	true	true
Error Message		error_message
Purchase Order Response	{ "Purchase Order Response": { "docnum": "0000000000199019", "status": "Success", "statusCode": "53", "status..." } }	
Status	success	status

Object can be further inspected by clicking on value.

Viewing response [object]



Raw Text

Formatted

```
{
  "Purchase Order Response": {
    "docnum": "0000000000199019",
    "status": "Success",
    "statusCode": "53",
    "statusDesc": "Inbound IDoc posted",
    "tid": "0A001E68852065BFE2F20003"
  }
}
```

Close

Step 5 – Validation on SAP Side

Display change material status tcode MM04

